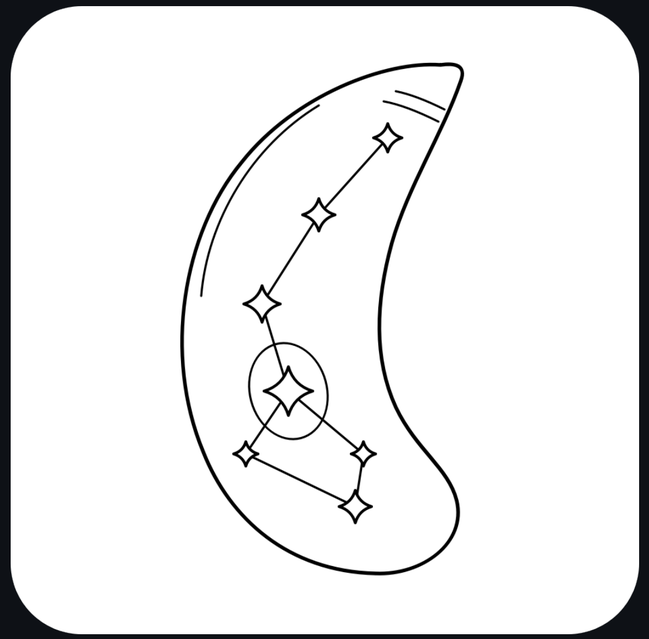
ParaScape

Explore what is known.
Evaluate what a model predicts.



A practical guide to the 0.43 release

Einar Olafsson · github.com/EinarOlafsson/parascape

Start with a gene. Keep the evidence.

Bring scattered measurements together without losing what they mean.

01

Collect

Published measurements, annotations and protein features.

02

Connect

Match gene identities and retain the source of each value.

03

Explore

Read evidence, compare relationships and inspect gaps.

04

Evaluate

Test predictions on held-out genes before interpreting them.

A model suggests a hypothesis. Its evidence and validation determine how useful it is.

Two organisms. Separate maps.

Offline gene tables and networks are included in the installation.

8,140

Toxoplasma gondii genes

5,720

Plasmodium falciparum genes

130

Registered datasets and computed layers

Coverage differs by assay and organism. These counts do not imply complete evidence for every gene.

Choose the task you came to do.

Explore a gene

Find an accession, symbol or product. Read measurements and their sources.

Predict a trait

Choose an observed outcome. Evaluate a model, then inspect unlabelled-gene hypotheses.

Compare a screen

Join a gene-level results table to the existing evidence. Keep unresolved IDs visible.

Open these guided workflows from the Tools menu.

The map is one view of the evidence.

One point represents one gene.

Position

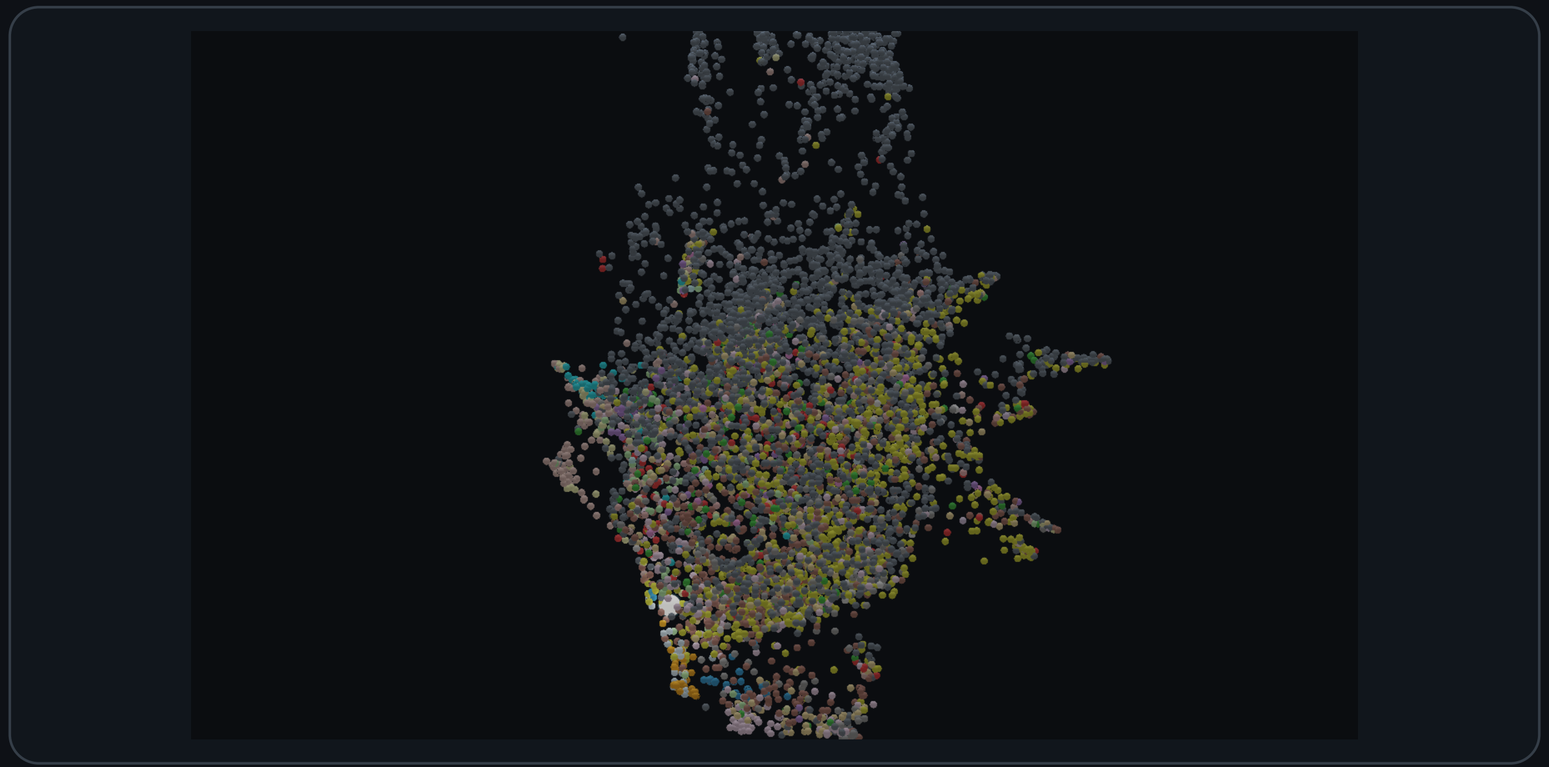
Similarity under the chosen feature recipe.

Colour

An annotation or measurement you select.

Lines

One explicitly selected relationship type.



Proximity alone does not establish function or a physical interaction.

Install and open ParaScape.

```
pip install parascape
parascape
```

Python 3.10+

Use a dedicated environment when possible.

Desktop display

The viewer needs a display and working OpenGL.

CPU included

CUDA is optional; browsing does not require a GPU.

Linux, macOS and Windows. Built tables are available offline.

Your first five minutes.

01

1 Choose

File > Species: select the organism.

02

2 Search

Tools > Explore a gene: enter an accession or product.

03

3 Inspect

Read the evidence and open its source links.

04

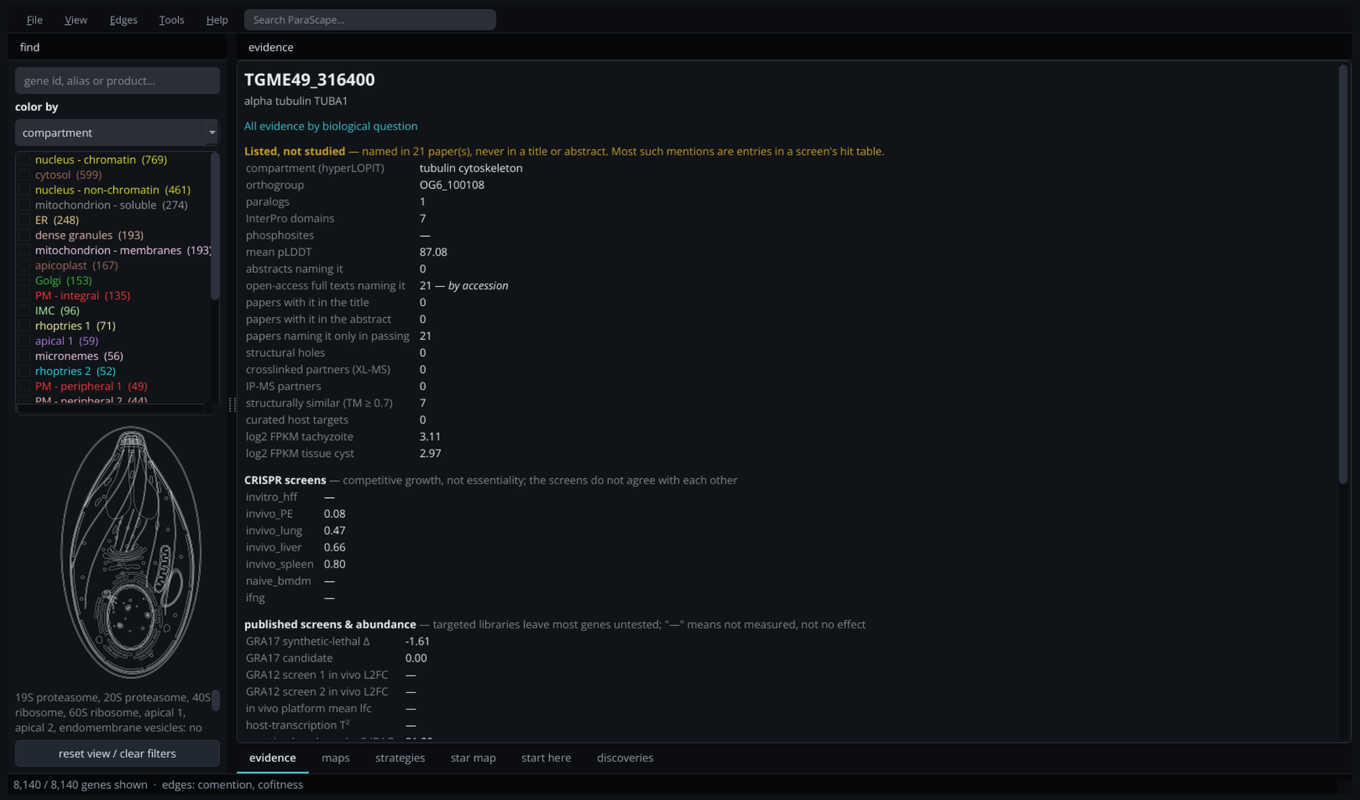
4 Compare

Change a colour or edge layer; keep the question specific.

Start with one familiar gene so you can check that the interpretation makes sense.

Find your way around the workspace.

Map controls on the left. The gene map in the centre. Evidence on the right.



Tools opens guided workflows and analysis docks. View controls appearance and interaction.

EXPLORE

Move the map. Keep your bearings.

Rotate and zoom

Left-drag in Navigate mode to rotate. Scroll to move closer or farther away.

Select a gene

Click a point to show its evidence. Search can take you directly to a gene.

Focus a category

Filter checked categories. Double-click a category to centre its genes.

Right-click the map for display options, export and reset.

Search by identity or description.

Tools > Explore a gene

Enter a query

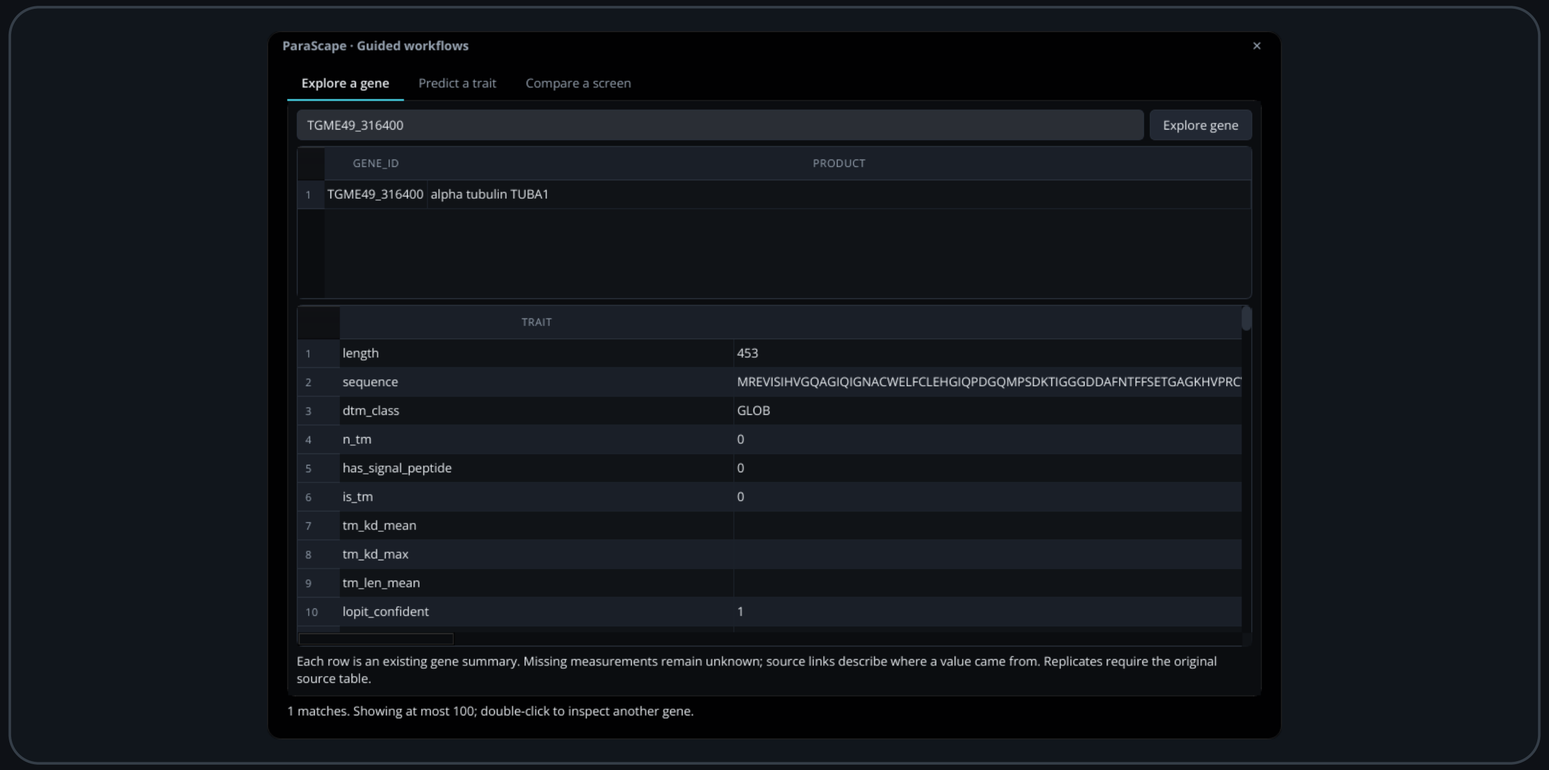
Use a gene accession, symbol or product text.

Check the match

Double-click a result to select the same gene on the map.

Read the table

Inspect available values, missingness and source evidence.



Shown here: an existing alpha-tubulin record, not a model-generated annotation.

EVIDENCE

Read the source, not just the value.

What was measured?

Distinguish expression, localization, abundance, sequence and interaction evidence.

In which context?

Check the organism, stage, condition and assay before comparing measurements.

How was it obtained?

Separate measured evidence, computed features, transferred annotations and predictions.

Legacy gene summaries cannot reconstruct replicate-level uncertainty that was never retained.

EVIDENCE

The full data catalogue stays one click away.

Evidence family	Examples	Read it as
Expression	Stage profiles and summaries	Condition-dependent measurements
Localization	Compartment annotations	Evidence with assay-specific coverage
Protein features	Sequence, domains, AF3 summaries	Measured or computed descriptors
Relationships	Coexpression and physical assays	Distinct types of connections
Literature	Mentions and reviewed assertions	Traceable source context

Each catalogue record links to its publication, source data or computed inputs.

EVIDENCE

Unknown is different from zero.

0

A measured value that really is zero

Unknown

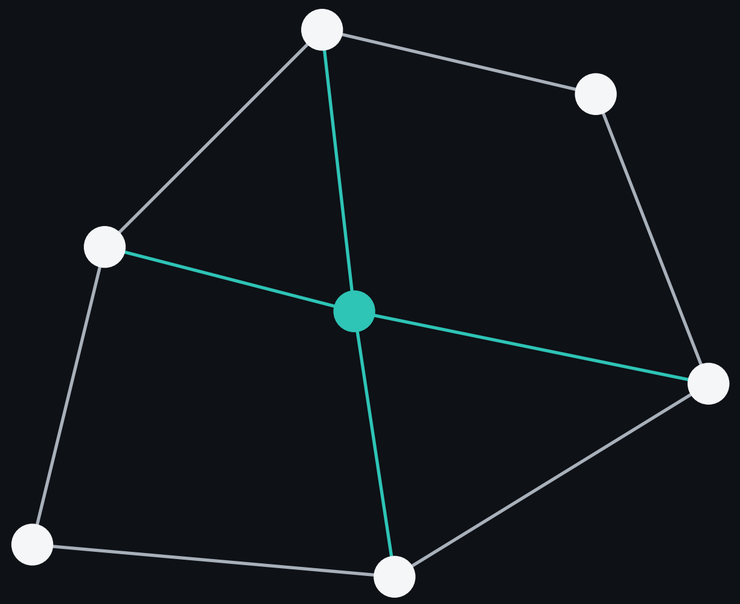
No usable value is available

No call

The model lacks enough support

A missing annotation is not a negative example. Grey marks unknown evidence.

Different lines answer different questions.



Illustrative network; line meaning comes from its layer.

The Edges controls keep these layers separate.

Similarity

Shared domains, orthogroups, expression or structure.

Physical evidence

Crosslink MS and pulldown assays have their own interpretation.

Literature context

Co-mention means appearing together in text; it is not proof of binding.

Inspect one connection layer at a time.

01

Select

Click a gene you want to inspect.

02

Choose

Turn on a relevant layer in Edges.

03

Read

Check neighbouring genes and the evidence behind the relation.

04

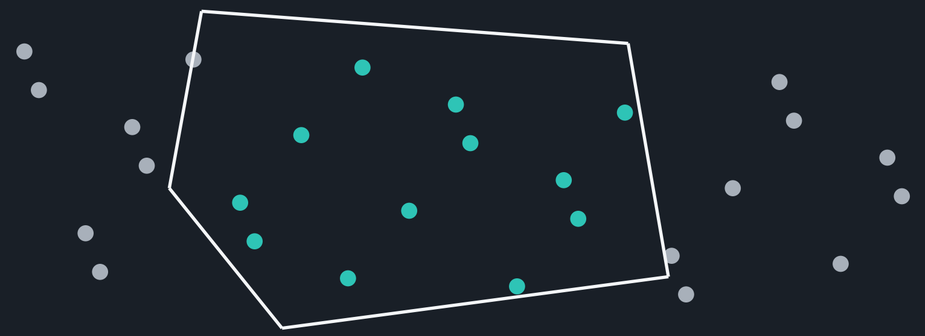
Compare

Switch the layer and see which relationships persist.

Draw all active edges is available, but dense networks are often easier to understand locally.

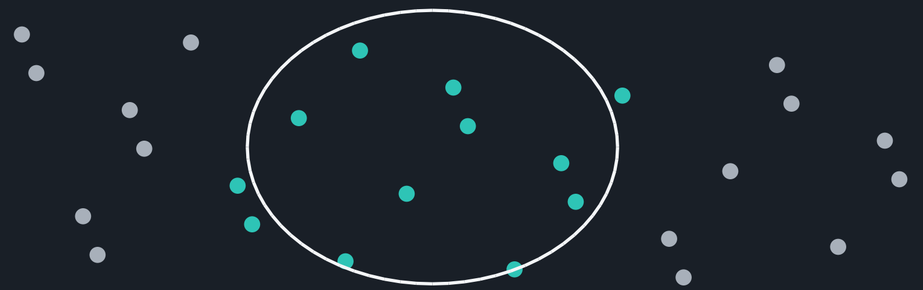
Select a group without changing the data.

Lasso: screen space



Draw an outline around visible positions. Points behind one another can be included.

Brush: 3D space



Select a spatial neighbourhood around the gene you press.

Choose View > Left mouse button > Select. Check the selected gene list before exporting.

Explore clusters and compare analyses.

Clusters

Group similar coordinates with clustering methods. Noise label -1 means unassigned, not a biological class.

Search and validation

Compare recipes with explicit targets and budgets. Confirm a search winner with independent evidence.

Saved results

Name embeddings and retain their recipes. Right-click result tables to export them; save selected gene lists.

Cluster labels describe the chosen inputs. They do not establish a shared biological function.

Change a map deliberately.

01

Features

Choose relevant measurement blocks.

02

Missingness

Decide how absent inputs are handled.

03

Scaling

Keep feature magnitudes and blocks comparable.

04

Embedding

Choose a method, dimensions and a seed.

Smaller UMAP neighbourhoods favour local structure. Lower minimum distance packs points more tightly.

Explore in 3D. Evaluate separately.

The displayed map

An exploratory view of selected inputs. Numeric localization confidence can be among them.

A held-out prediction

A separate fitted model with target exclusions, training-only transforms and untouched evaluation groups.

The guided predictor does not classify the opening map.

Set up a prediction in one place.

Tools > Predict a trait

Outcome and type

Choose the observed target and classification or regression.

Method and groups

Start with a baseline. Keep related genes together with orthogroup.

Run and inspect

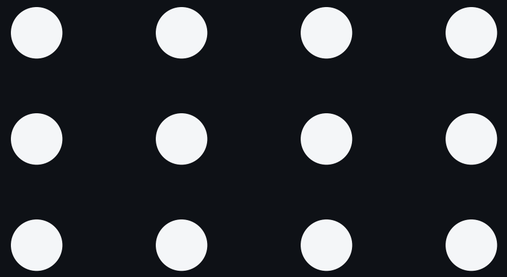
Evaluate and predict. Read held-out metrics before the candidate table.



The Jobs panel runs the work. Stop requests cancellation between model fits.

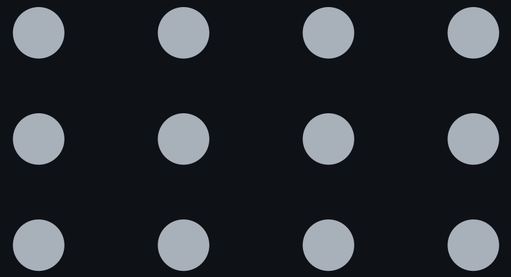
Keep evaluation genes out of fitting.

Fit



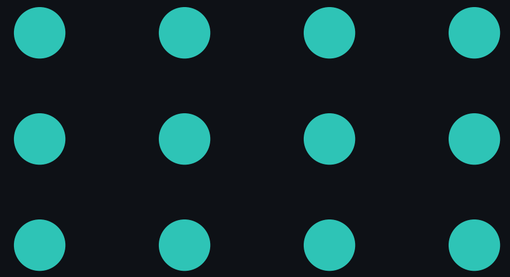
Learn features and model parameters.

Calibrate



Adjust probabilities or residual intervals on separate groups.

Evaluate



Score predictions on groups withheld from both steps.

Circles represent genes. Related genes stay within one group.

Outer folds rotate the evaluation groups. Grouped validation is harder and more informative than mixing relatives.

Compare simple methods first.

Method	Useful starting question	Keep in mind
Prior / linear	Is there signal beyond a simple baseline?	Always retain a baseline
Boosted trees	Do nonlinear feature combinations help?	Use the same evaluation groups
Feature / PCA neighbours	Do similar measured genes agree?	Distance depends on preprocessing
UMAP / masked factors	Does a reduced representation help?	A compact map can lose signal
Weighted networks	Which fixed graph layers support labels?	Transductive; disconnected genes abstain

No method is automatically best because it is more complex.

Read performance and coverage together.

Accuracy

The fraction of evaluation genes called correctly. Unsupported calls count as misses.

Macro F1

Balances precision and recall across classes, so common labels do not dominate the score.

Coverage

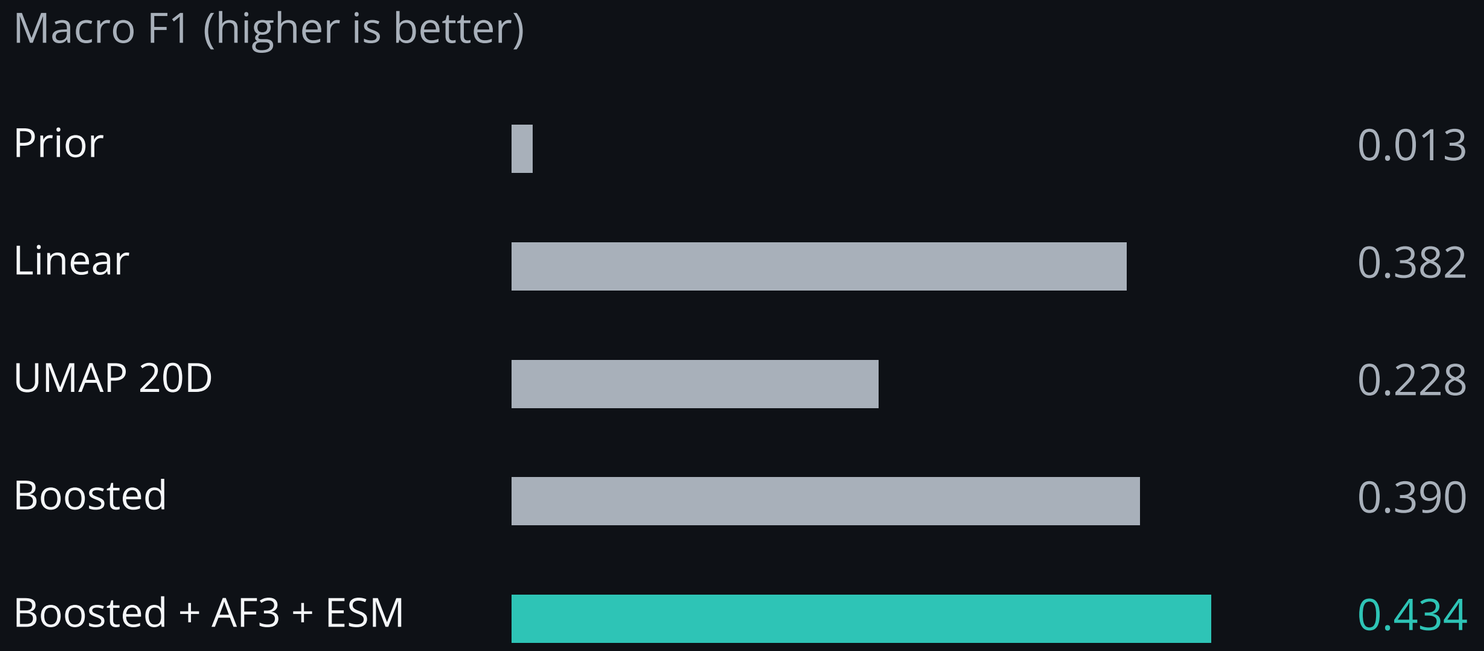
The fraction with a supported call. Selective accuracy describes only that supported subset.

Also inspect per-class precision, recall, average precision and calibration.

RESULTS

What worked in the localization benchmark?

3,827 measured genes · 26 classes · three orthogroup-held-out folds



0.434 macro F1

Boosted trees with published, AF3 and ESM inputs.

56.2% accuracy

Better than the compared UMAP-neighbour models.

Important limit

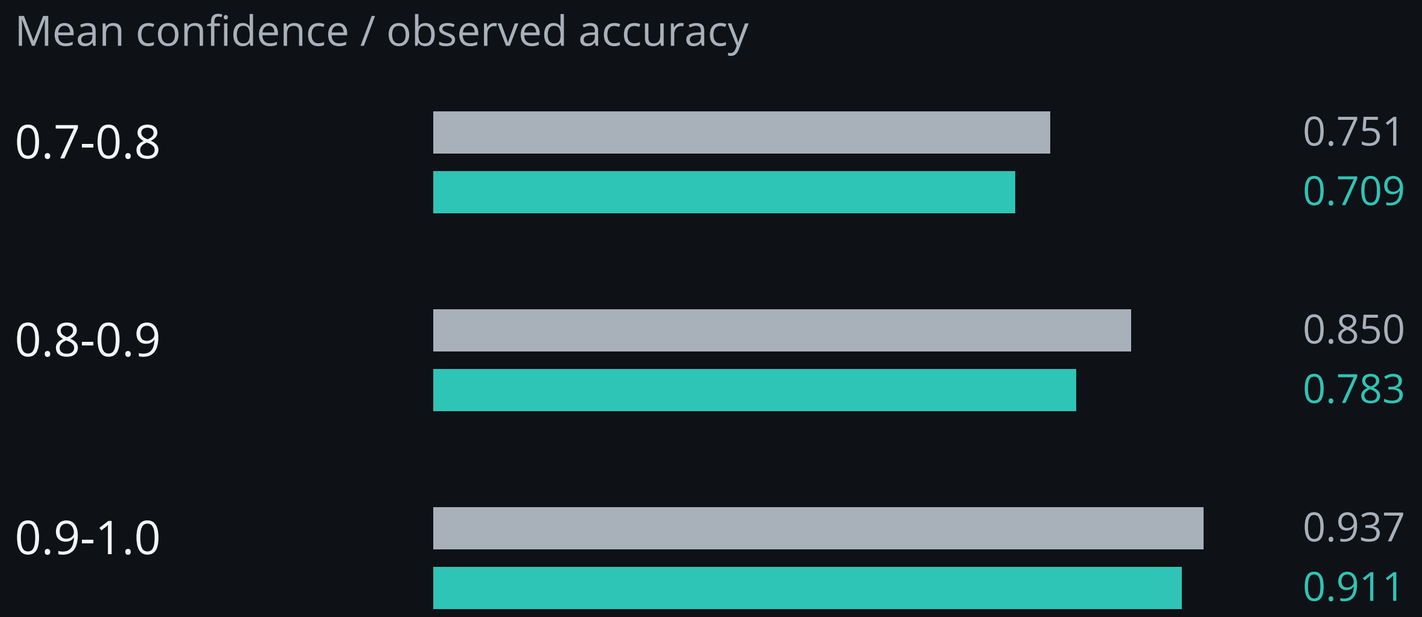
Several rare classes still have zero top-class recall.

Development cross-validation, not prospective validation of new annotations.

RESULTS

A confident score can still be wrong.

Combined boosted model: observed accuracy in held-out confidence bins



Check calibration

Temperature scaling uses separate groups when enough labels exist.

Read the status

The app explicitly marks runs that could not be calibrated.

Use abstention

A probability cutoff can withhold uncertain calls; it is not a guarantee.

The 0.8-0.9 bin averaged 0.850 confidence but 0.783 accuracy across 465 genes.

PROTEIN EVIDENCE

Add sequence evidence with one checkbox.

8,064 / 8,140

Toxoplasma genes with this feature view



99.1% coverage in the bundled release

8,064 proteins

Bundled frozen ESM-2 representations.

320 features

A numerical view of protein sequence, separate from experimental measurements.

No download to use

Enable Include frozen protein sequence features in the guided predictor.

Regenerating representations needs the optional sequence extra. Benchmark their value rather than assuming it.

Your AF3 collection becomes traceable evidence.

1,210 / 8,140

Toxoplasma genes with this feature view



14.9% coverage in the bundled release

1,210 matched genes

Exact sequence or uniquely placed fragment matches in the available local collection.

Nine summary features

Confidence, sequence coverage and descriptive geometry.

Coordinates stay local

The package includes features and provenance, not the private model files.

This is the verified subset in the synced folders, not the total number of proteins ever folded.

Connect local folds to the structure viewer.

```
pip install 'parascape[structures]'

parascape-index-structures /path/to/models
```

Index once

Build the local structure catalogue from your model folders.

Select a gene

The structure viewer checks the local index before AlphaFold DB.

Read confidence

A predicted fold and its confidence are not experimental validation.

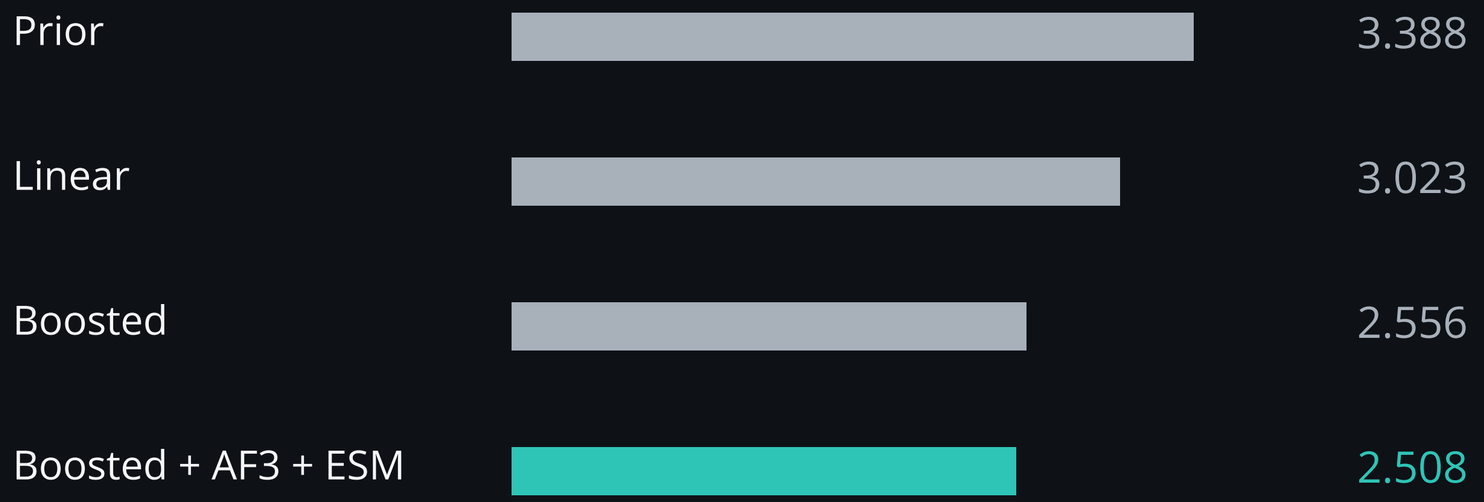
Recreate the index if the drive moves. Full models take precedence over fragments.

RESULTS

Continuous traits stay continuous.

Protein abundance: 748 measured genes · grouped held-out evaluation

RMSE in log2 units (lower is better)



Choose regression

Use Continuous measurement instead of converting an outcome into classes.

Read error units

RMSE here is measured in log2 abundance units; lower is better.

Small added gain

The combined-feature improvement interval includes no improvement.

Nominal 90% intervals covered 94.9%, but their mean width was 9.94 log2 units.

RESULTS

Network coverage is not prediction quality.

87.3% supported

Combining domain, expression and structural layers reaches more evaluation genes.

0.195 macro F1

The combined fixed network remains well below the compared feature models.

The graph includes evaluation nodes: report these results as transductive, not transfer to unseen nodes.

Compare a screen without losing unmatched rows.

Tools > Compare a screen

Open a CSV or TSV

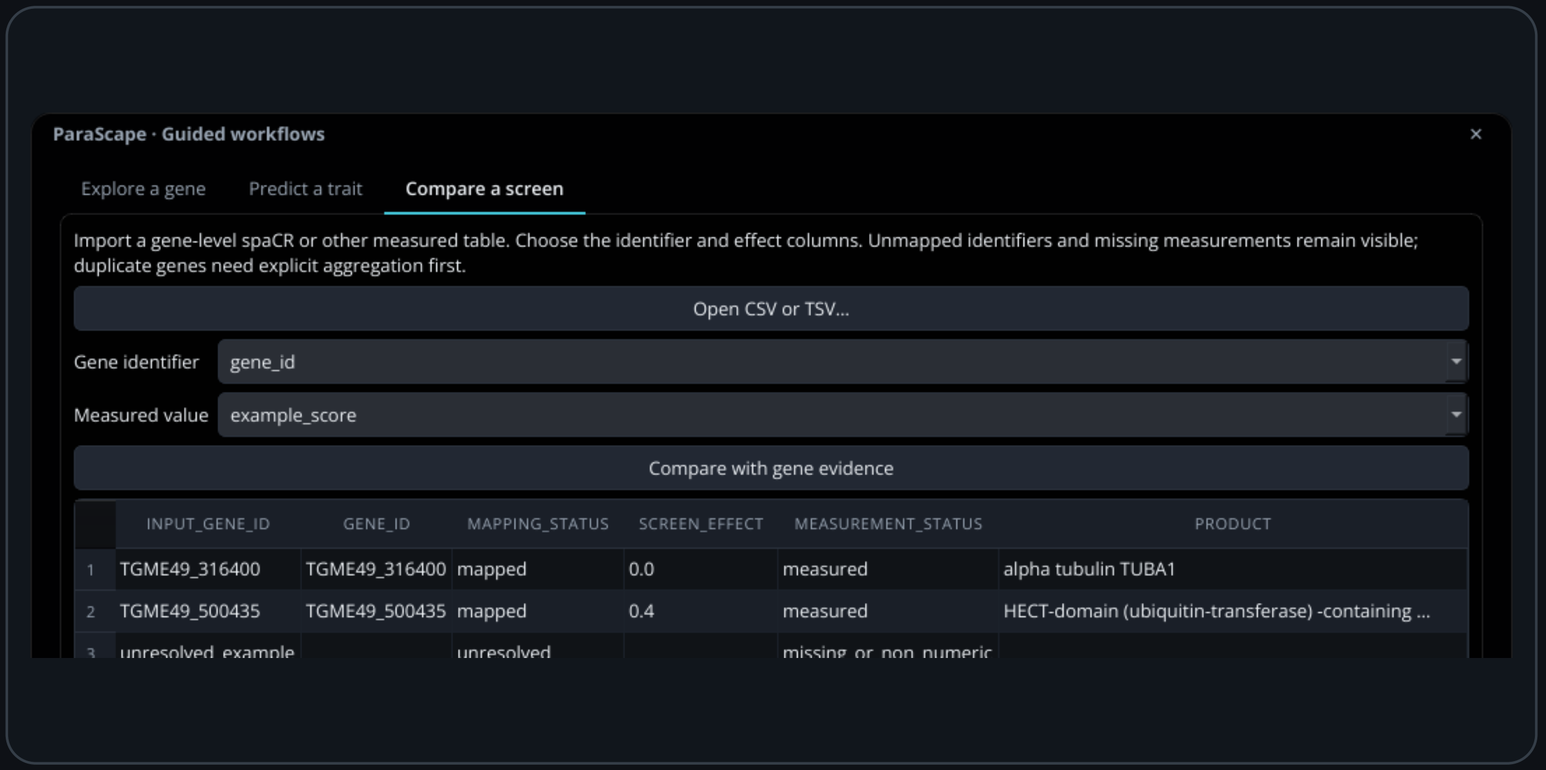
Use a gene-level table and choose its identifier and value columns.

Inspect the join

Mapped and unresolved IDs remain visible. Measured zero stays zero.

Export the comparison

Keep the table and source-file hash together.



Screenshot values are illustrative. They are not biological screen results.

SCREEN WORKFLOW

Move from spaCR results to gene evidence.

01

spaCR

Generate gene-level measurement or effect estimates.

02

CSV / TSV

Export identifiers and the measured value you want to compare.

03

ParaScape

Open Compare a screen and inspect the identifier mapping.

04

Evidence

Read the existing annotations and source context alongside the result.

The spaCR launcher opens ParaScape in a separate environment. It does not automatically transfer results.

SCREEN WORKFLOW

Add measurements to a Toxoplasma session.

File > Import data

The advanced importer accepts CSV, TSV, Excel or Parquet for Toxoplasma accessions.

Check the preview

Choose a useful column prefix and explicitly handle duplicates and missing values.

Rebuild deliberately

Imported values do not change map positions until you include them in a new embedding.

Advanced table import is Toxoplasma-only. The guided comparison supports the currently open organism.

REPRODUCIBILITY

Export the evidence behind a result.

01

predictions.csv

Separate held-out evaluations from unknown-gene hypotheses.

02

per_class.csv

Inspect performance and support for each class.

03

run.json

Retain settings, input identity, split genes, excluded features and versions.

Use Export full run. A ranked list alone cannot reproduce an analysis.

PYTHON API

Use the same methods from Python.

```
import pandas as pd
from parascape import paths
from parascape.prediction import TaskSpec, run

nodes = pd.read_parquet(
paths.cache_file('nodes.parquet'))
result = run(nodes, TaskSpec('compartment'))
result.save('localization_run')
```

Headless analysis

Read bundled tables and run methods without opening the desktop.

One contract

The API uses the same evaluation and export rules as the guided workflow.

The generated module reference documents functions, arguments and return values.

EVIDENCE API

Keep claims traceable and reviewable.

Observation records

Store entity, trait, value, units, context, source, uncertainty and missingness.

Literature assertions

Retain the source passage and negation. Extracted claims begin pending review.

Explicit limits

The API supports review records; the desktop has no bulk literature-curation interface.

A reviewed interpretation of a passage is not experimental proof. Conflicting observations remain separate.

PRACTICAL USE

Run work in the background. Keep control.

Jobs and Stop

Watch progress in the Jobs panel. Cancellation happens between model fits.

Optional GPU

CPU analysis is the default. NVIDIA CUDA analysis uses the gpu extra on supported Linux systems.

Logs and help

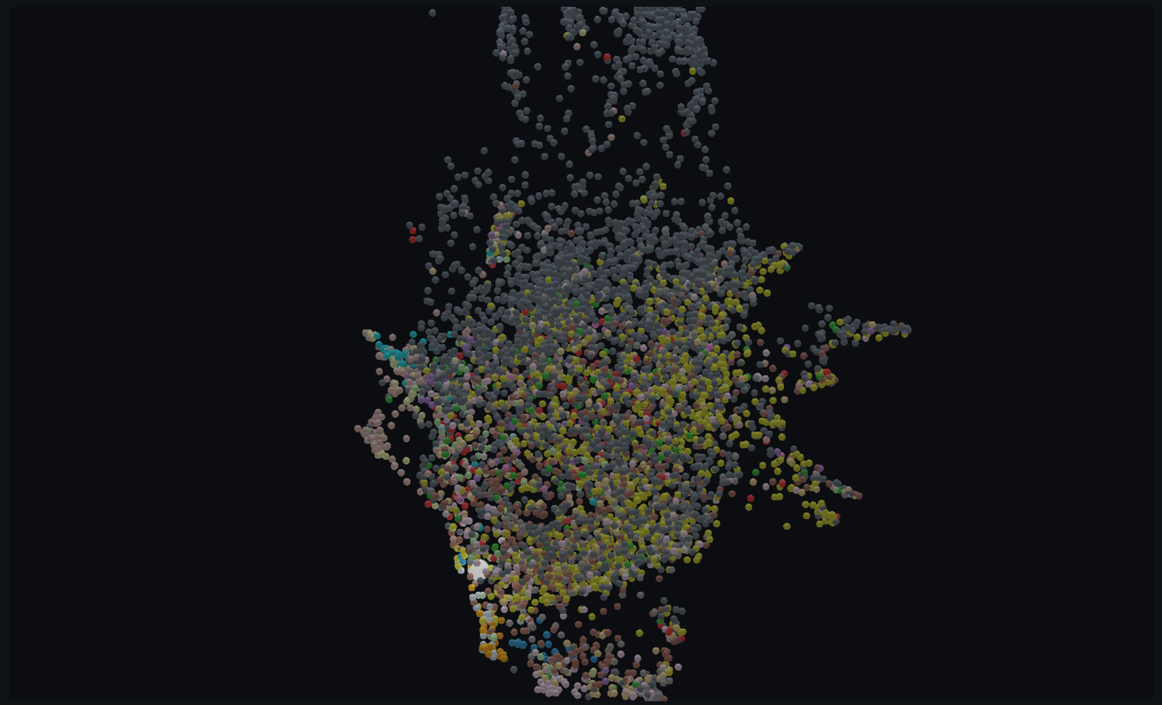
Hover over settings. Enable file logging in Preferences when investigating a problem.

Offline browsing works immediately. Remote structures, publications and optional chat can require network access.

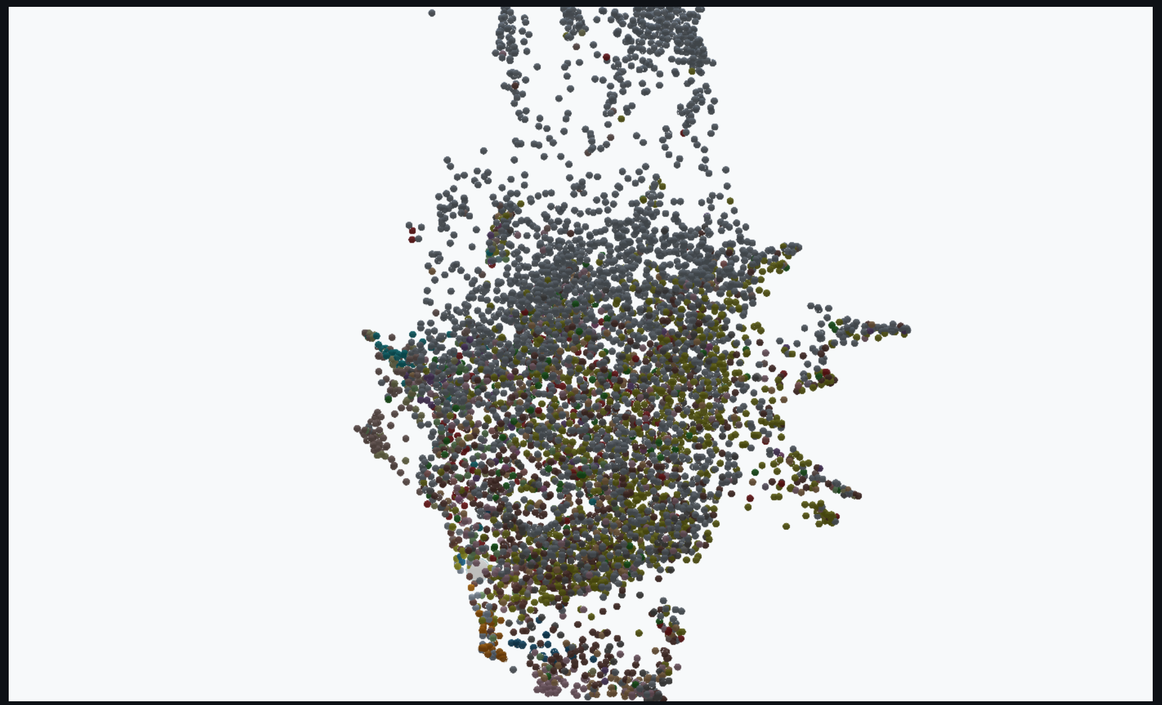
APPEARANCE

Choose a readable view.

View > Preferences: theme, colours, point rendering, lighting and text size.



Dark theme



Light theme

Lighting changes presentation only. Use neutral lighting when comparing measurement colours.

GOOD PRACTICE

Before interpreting a prediction, check four things.

01

Identity

Are the gene mapping, organism and assay context correct?

02

Evidence

Which values are measured, computed, missing or transferred?

03

Validation

Were target-related inputs excluded and related genes held together?

04

Uncertainty

Do rare classes, calibration and coverage support the intended interpretation?

Model comparisons guide exploration. Independent evidence is needed to establish a new biological claim.

START HERE

Open a gene. Ask a precise question.

```
pip install parascape
```

User guide

Open resource >

Guided workflows

Open resource >

Data sources

Open resource >

Python API

Open resource >

Benchmarks

Open resource >

Issues and source

Open resource >

ParaScape · Einar Olafsson · Open-source software, with original data sources credited.